Dataset Validation

Surgical Scene Semantic Segmentation

Dataset: sarrarp50

Classes: 10 | Videos: 54 | Frames: 16,295

Class presence is analysed at three levels - frames, pixels and videos - on the complete dataset (no sampling), globally and for every split. Video coverage highlights class imbalance that frame and pixel counts alone can hide.

1. Executive Summary

Dataset: sarrarp50

Dataset

- 10 classes, 54 videos, 16,295 frames.
- train: 13,043 frames, 44 videos
- val: 3,252 frames, 10 videos
- Ignored (out-of-range) pixels: 0

Class imbalance warnings (global, thresholds: <= 3 videos, >= 50% pixels in one video)

- No class is rare or concentrated in few videos.

Classes missing from a split

- Every class appears in every split.

Dominant / smallest pixel share (global)

- Dominant: Background (78.95%)
- Smallest: Clamps (0.1771%)

2. Dataset Overview

Dataset: sarrarp50

Split	Frames	Videos
train	13,043	44
val	3,252	10
global	16,295	54

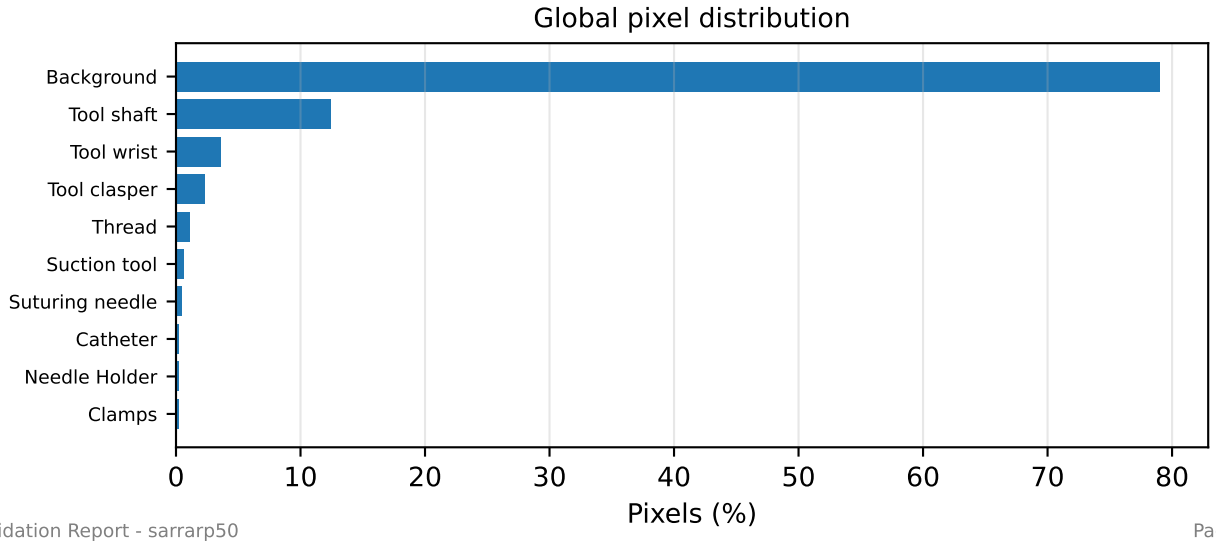
Classes

```
0  Background
1  Tool clasper
2  Tool wrist
3  Tool shaft
4  Suturing needle
5  Thread
6  Suction tool
7  Needle Holder
8  Clamps
9  Catheter
```

3. Global Statistics

All splits together - rows highlighted = class imbalance warning (part 1)

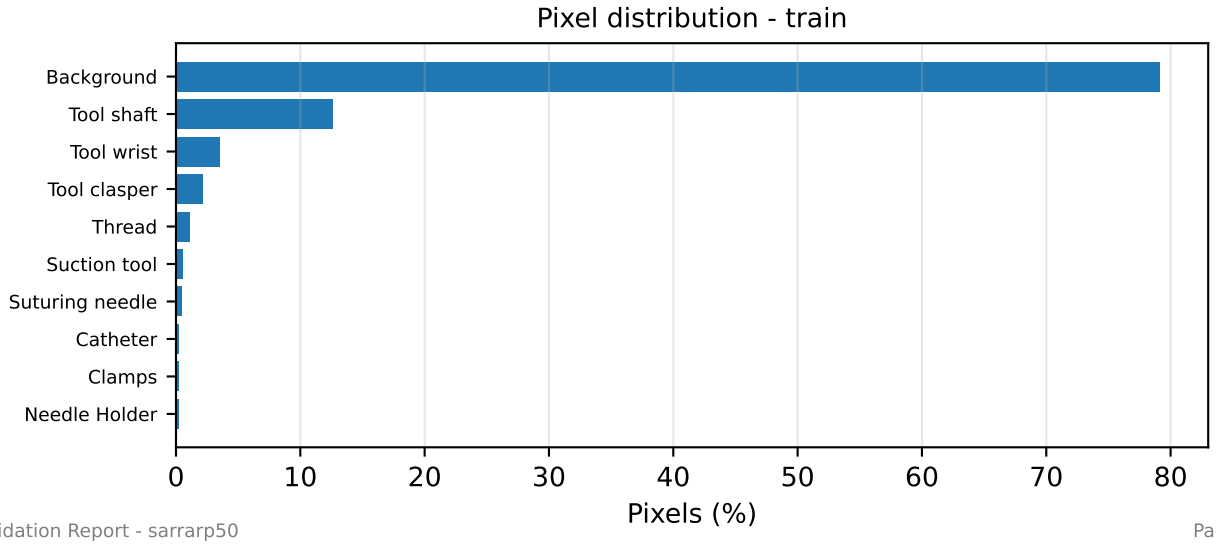
Class	Frames	Frames %	Pixels	Pixels %	Videos
Background	16,295	100.00	26,678,120,442	78.9543	54/54
Tool clasper	15,701	96.35	775,009,051	2.2937	54/54
Tool wrist	15,494	95.08	1,205,213,603	3.5668	54/54
Tool shaft	14,932	91.64	4,197,536,557	12.4227	54/54
Suturing needle	12,670	77.75	156,204,401	0.4623	54/54
Thread	14,874	91.28	365,784,043	1.0825	54/54
Suction tool	1,942	11.92	212,592,050	0.6292	49/54
Needle Holder	458	2.81	61,403,884	0.1817	54/54
Clamps	10,927	67.06	59,833,614	0.1771	51/54
Catheter	7,458	45.77	77,614,355	0.2297	33/54



4. Split Statistics - train

13,043 frames, 44 videos - highlighted = class in <= 3 videos (part 1)

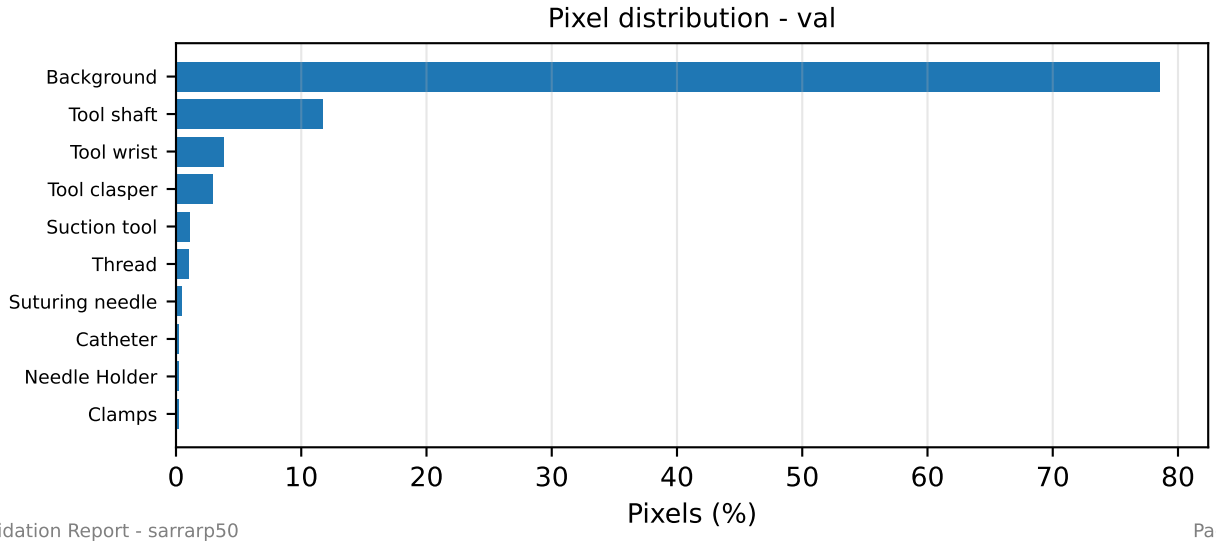
Class	Frames	Frames %	Pixels	Pixels %	Videos
Background	13,043	100.00	21,385,467,795	79.0708	44/44
Tool clasper	12,567	96.35	580,043,002	2.1447	44/44
Tool wrist	12,401	95.08	951,629,304	3.5186	44/44
Tool shaft	12,003	92.03	3,409,308,790	12.6056	44/44
Suturing needle	10,037	76.95	123,939,049	0.4583	44/44
Thread	11,898	91.22	297,687,870	1.1007	44/44
Suction tool	1,359	10.42	138,590,373	0.5124	41/44
Needle Holder	381	2.92	48,120,435	0.1779	44/44
Clamps	9,011	69.09	48,676,302	0.1800	43/44
Catheter	6,162	47.24	62,501,880	0.2311	28/44



4. Split Statistics - val

3,252 frames, 10 videos - highlighted = class in <= 3 videos (part 1)

Class	Frames	Frames %	Pixels	Pixels %	Videos
Background	3,252	100.00	5,292,652,647	78.4870	10/10
Tool clasper	3,134	96.37	194,966,049	2.8912	10/10
Tool wrist	3,093	95.11	253,584,299	3.7605	10/10
Tool shaft	2,929	90.07	788,227,767	11.6890	10/10
Suturing needle	2,633	80.97	32,265,352	0.4785	10/10
Thread	2,976	91.51	68,096,173	1.0098	10/10
Suction tool	583	17.93	74,001,677	1.0974	8/10
Needle Holder	77	2.37	13,283,449	0.1970	10/10
Clamps	1,916	58.92	11,157,312	0.1655	8/10
Catheter	1,296	39.85	15,112,475	0.2241	5/10



5. Video Coverage

In how many (and which) videos each class appears - global. Top share = % of the class pixels in its dominant video

Class	Videos	Videos %	Top video	Top share %	Which videos
Background	54/54	100	video_06	4	video_01, video_02, video_03, video_04, video_05, ... (+49)
Tool clasper	54/54	100	video_03	5	video_01, video_02, video_03, video_04, video_05, ... (+49)
Tool wrist	54/54	100	video_31	5	video_01, video_02, video_03, video_04, video_05, ... (+49)
Tool shaft	54/54	100	video_06	5	video_01, video_02, video_03, video_04, video_05, ... (+49)
Suturing needle	54/54	100	video_49	7	video_01, video_02, video_03, video_04, video_05, ... (+49)
Thread	54/54	100	video_01	7	video_01, video_02, video_03, video_04, video_05, ... (+49)
Suction tool	49/54	91	video_49	22	video_01, video_02, video_03, video_04, video_05, ... (+44)
Needle Holder	54/54	100	video_14	6	video_01, video_02, video_03, video_04, video_05, ... (+49)
Clamps	51/54	94	video_06	12	video_01, video_02, video_03, video_04, video_05, ... (+46)
Catheter	33/54	61	video_31	18	video_01, video_03, video_05, video_06, video_09, ... (+28)

5. Video Coverage - train

44 videos in this split (part 1)

Class	Videos	Top video	Top share %	Which videos
Background	44/44	video_06	5	video_01, video_02, video_03, video_04, video_05, ... (+39)
Tool clasper	44/44	video_03	7	video_01, video_02, video_03, video_04, video_05, ... (+39)
Tool wrist	44/44	video_31	6	video_01, video_02, video_03, video_04, video_05, ... (+39)
Tool shaft	44/44	video_06	6	video_01, video_02, video_03, video_04, video_05, ... (+39)
Suturing needle	44/44	video_01	6	video_01, video_02, video_03, video_04, video_05, ... (+39)
Thread	44/44	video_01	9	video_01, video_02, video_03, video_04, video_05, ... (+39)
Suction tool	41/44	video_32	11	video_01, video_02, video_03, video_04, video_05, ... (+36)
Needle Holder	44/44	video_14	8	video_01, video_02, video_03, video_04, video_05, ... (+39)
Clamps	43/44	video_06	15	video_01, video_02, video_03, video_04, video_05, ... (+38)
Catheter	28/44	video_31	23	video_01, video_03, video_05, video_06, video_09, ... (+23)

5. Video Coverage - val

10 videos in this split (part 1)

Class	Videos	Top video	Top share %	Which videos
Background	10/10	video_49	21	video_41, video_42, video_43, video_44, video_45, ... (+5)
Tool clasper	10/10	video_42	18	video_41, video_42, video_43, video_44, video_45, ... (+5)
Tool wrist	10/10	video_49	22	video_41, video_42, video_43, video_44, video_45, ... (+5)
Tool shaft	10/10	video_46	23	video_41, video_42, video_43, video_44, video_45, ... (+5)
Suturing needle	10/10	video_49	33	video_41, video_42, video_43, video_44, video_45, ... (+5)
Thread	10/10	video_46	23	video_41, video_42, video_43, video_44, video_45, ... (+5)
Suction tool	8/10	video_49	64	video_42, video_43, video_45, video_46, video_47, ... (+3)
Needle Holder	10/10	video_46	14	video_41, video_42, video_43, video_44, video_45, ... (+5)
Clamps	8/10	video_49	47	video_42, video_44, video_45, video_46, video_47, ... (+3)
Catheter	5/10	video_46	42	video_44, video_45, video_46, video_47, video_50

5. Video x Class table (frames and pixels)

Rows = videos, columns = classes; under each class: frames containing the class | class pixels.
Red columns = rare class (<= 3 videos or >= 50% of pixels in one video). Grey 0 = class absent from the video. (classes part 1/2, videos part 1/3)

	Background		Tool clasper		Tool wrist		Tool shaft		Suturing needle		Thread	
Video [split]	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**
video_01 [train]	563	899.91M	558	25.56M	551	56.29M	514	138.68M	479	7.40M	559	27.22M
video_02 [train]	330	585.92M	275	10.46M	268	17.07M	250	55.18M	229	3.24M	273	7.08M
video_03 [train]	522	876.11M	521	38.59M	520	39.80M	499	101.86M	435	3.26M	499	8.90M
video_04 [train]	131	225.36M	126	11.03M	106	7.93M	93	21.59M	91	973.8k	107	2.65M
video_05 [train]	272	420.85M	261	37.01M	246	26.19M	223	53.21M	192	3.33M	256	9.32M
video_06 [train]	706	1145.36M	684	21.03M	686	41.85M	679	221.18M	560	6.39M	679	12.48M
video_07 [train]	272	439.50M	260	14.83M	258	28.67M	238	60.04M	217	4.71M	244	6.65M
video_08 [train]	148	246.08M	148	4.07M	148	7.99M	146	38.05M	99	978.5k	118	2.10M
video_09 [train]	264	415.48M	236	6.12M	240	22.81M	236	87.59M	167	2.53M	221	7.87M
video_10 [train]	128	211.95M	128	4.66M	125	10.53M	118	32.91M	101	1.36M	124	3.11M
video_11_1 [train]	237	396.33M	232	13.56M	223	21.02M	204	47.48M	206	3.32M	219	6.93M
video_11_2 [train]	135	213.96M	135	7.10M	135	15.86M	129	33.31M	130	2.22M	133	2.08M
video_12 [train]	132	218.64M	132	5.25M	132	12.57M	128	31.49M	85	846.4k	132	3.14M
video_13 [train]	273	462.76M	251	30.31M	238	20.70M	214	33.57M	198	2.47M	235	4.63M
video_14 [train]	282	497.48M	261	23.45M	230	18.92M	182	30.95M	203	3.13M	232	4.43M
video_15_1 [train]	180	306.79M	159	5.51M	154	11.63M	166	43.11M	120	1.55M	149	3.76M
video_15_2 [train]	132	216.48M	132	4.40M	130	10.13M	130	36.61M	117	1.28M	125	1.92M
video_16 [train]	385	638.61M	376	11.71M	366	21.10M	367	109.00M	247	3.27M	366	12.12M
video_17_1 [train]	281	453.05M	281	8.89M	280	13.26M	268	95.75M	250	2.45M	260	5.54M
video_17_2 [train]	132	215.95M	132	8.11M	132	14.78M	131	28.20M	124	1.37M	129	2.27M
video_18 [train]	545	867.84M	500	15.57M	500	28.98M	485	188.32M	383	3.72M	482	11.95M
video_19 [train]	267	438.40M	253	14.71M	250	30.71M	223	55.99M	243	4.46M	248	6.85M
video_20 [train]	290	461.31M	285	9.42M	285	24.73M	284	90.17M	201	2.80M	280	7.60M
video_21 [train]	455	700.09M	404	17.01M	416	43.10M	397	143.94M	303	6.23M	404	12.93M
video_22 [train]	383	658.37M	364	8.76M	359	16.81M	348	97.06M	291	2.48M	358	8.78M
video_23 [train]	257	425.30M	246	8.07M	244	16.42M	251	71.57M	212	1.92M	242	7.28M

5. Video x Class table (frames and pixels)

Rows = videos, columns = classes; under each class: frames containing the class | class pixels.
Red columns = rare class (<= 3 videos or >= 50% of pixels in one video). Grey 0 = class absent from the video. (classes part 1/2, videos part 2/3)

	Background		Tool clasper		Tool wrist		Tool shaft		Suturing needle		Thread	
Video [split]	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**
video_24 [train]	250	407.70M	236	8.89M	238	18.18M	232	68.90M	211	2.60M	224	7.06M
video_25 [train]	154	255.86M	153	9.36M	153	18.70M	150	27.89M	141	1.01M	147	2.54M
video_26 [train]	164	274.38M	164	11.67M	156	12.19M	157	35.10M	141	1.33M	151	3.71M
video_27 [train]	188	319.07M	185	11.30M	184	13.57M	177	36.68M	120	1.18M	126	2.67M
video_28 [train]	182	291.76M	181	5.78M	179	11.97M	170	58.26M	176	2.11M	180	3.38M
video_29_1 [train]	258	419.19M	252	8.38M	251	17.47M	246	77.80M	231	2.93M	240	6.24M
video_29_2 [train]	233	371.44M	232	7.41M	232	16.57M	228	70.35M	211	3.09M	230	3.85M
video_30 [train]	574	911.55M	574	23.18M	569	38.36M	563	189.58M	516	5.91M	562	12.27M
video_31 [train]	565	880.63M	535	24.67M	536	58.32M	538	166.41M	324	5.02M	471	11.08M
video_32 [train]	490	792.30M	488	12.62M	465	26.95M	462	147.66M	322	3.26M	422	11.61M
video_33 [train]	158	272.03M	158	7.11M	158	11.85M	158	31.96M	148	628.5k	157	2.00M
video_34 [train]	101	175.07M	101	8.95M	97	8.07M	82	14.46M	75	622.1k	94	1.39M
video_35 [train]	504	826.62M	499	23.42M	497	41.81M	485	129.19M	397	5.83M	463	13.82M
video_36 [train]	339	557.69M	338	17.18M	338	23.77M	324	81.63M	262	3.07M	316	9.05M
video_37 [train]	360	602.97M	342	7.92M	342	12.99M	357	113.64M	312	2.82M	325	4.33M
video_38 [train]	306	523.37M	286	17.81M	275	22.80M	265	61.46M	247	2.53M	252	5.05M
video_39 [train]	239	392.36M	228	5.76M	234	12.78M	231	72.63M	145	1.50M	227	3.37M
video_40 [train]	276	473.59M	275	3.43M	275	5.43M	275	78.89M	175	837.8k	237	4.66M
video_41 [val]	238	383.69M	237	24.02M	236	25.16M	216	54.23M	184	2.44M	217	3.02M
video_42 [val]	358	545.84M	358	35.19M	358	41.69M	339	96.92M	289	2.99M	341	9.53M
video_43 [val]	198	328.10M	192	17.21M	182	12.06M	177	42.90M	153	1.73M	184	4.77M
video_44 [val]	143	234.51M	133	22.98M	122	11.72M	91	19.81M	101	1.74M	119	3.56M
video_45 [val]	117	194.86M	113	13.17M	109	12.31M	101	15.89M	96	779.1k	104	1.86M
video_46 [val]	633	1041.18M	601	18.63M	593	36.87M	578	181.45M	442	3.75M	572	15.35M
video_47 [val]	334	551.29M	332	22.77M	331	28.52M	325	73.02M	303	2.94M	325	7.42M
video_48 [val]	301	497.48M	297	10.76M	298	21.69M	290	74.13M	266	3.75M	258	5.47M

5. Video x Class table (frames and pixels)

Rows = videos, columns = classes; under each class: frames containing the class | class pixels.
Red columns = rare class (<= 3 videos or >= 50% of pixels in one video). Grey 0 = class absent from the video. (classes part 1/2, videos part 3/3)

	Background		Tool clasper		Tool wrist		Tool shaft		Suturing needle		Thread	
Video [split]	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**
video_49 [val]	687	1116.62M	642	23.74M	634	54.58M	581	151.58M	588	10.61M	632	12.99M
video_50 [val]	243	399.08M	229	6.49M	230	8.98M	231	78.29M	211	1.54M	224	4.13M
TOTAL	**16,295**	**26678.12M**	**15,701**	**775.01M**	**15,494**	**1205.21M**	**14,932**	**4197.54M**	**12,670**	**156.20M**	**14,874**	**365.78M**

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Red columns = rare class (<= 3 videos or >= 50% of pixels in one video). Grey 0 = class absent from the video. (classes part 2/2, videos part 1/3)

	Suction tool		Needle Holder		Clamps		Catheter	
Video [split]	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**
video_01 [train]	25	3.68M	13	2.07M	523	3.53M	469	3.11M
video_02 [train]	6	288.5k	20	2.54M	217	2.51M	0	0
video_03 [train]	65	11.63M	4	450.3k	495	363.8k	284	1.45M
video_04 [train]	6	573.4k	8	1.40M	13	127.3k	0	0
video_05 [train]	55	5.28M	10	1.68M	263	5.02M	219	2.12M
video_06 [train]	67	3.84M	26	2.70M	686	7.31M	412	1.83M
video_07 [train]	96	8.27M	4	554.9k	232	803.7k	0	0
video_08 [train]	68	6.22M	11	1.20M	79	204.3k	0	0
video_09 [train]	17	2.45M	8	1.10M	75	77.5k	195	1.40M
video_10 [train]	3	181.9k	5	354.8k	127	363.5k	0	0
video_11_1 [train]	25	2.29M	4	323.3k	43	116.7k	15	67.8k
video_11_2 [train]	16	3.54M	9	1.27M	28	48.3k	73	543.1k
video_12 [train]	5	148.4k	5	349.5k	131	873.0k	123	414.1k
video_13 [train]	73	9.76M	8	1.89M	3	4,473	0	0
video_14 [train]	38	2.77M	13	3.62M	1	950	0	0
video_15_1 [train]	7	332.4k	3	170.7k	144	385.9k	0	0
video_15_2 [train]	3	283.5k	6	1.33M	100	434.5k	117	856.7k
video_16 [train]	2	90.0k	22	1.87M	218	564.7k	0	0
video_17_1 [train]	13	1.91M	8	878.5k	274	919.6k	9	24.6k
video_17_2 [train]	12	1.48M	4	701.9k	127	206.9k	122	641.5k
video_18 [train]	50	4.94M	11	1.38M	504	3.13M	461	4.28M
video_19 [train]	21	1.95M	2	307.8k	50	47.6k	19	215.2k
video_20 [train]	0	0	8	1.41M	214	289.7k	272	3.60M
video_21 [train]	50	11.33M	15	1.44M	283	2.32M	379	5.09M
video_22 [train]	13	1.33M	4	588.1k	0	0	0	0
video_23 [train]	15	1.65M	7	646.5k	186	45.4k	0	0

5. Video x Class table (frames and pixels)

Rows = videos, columns = classes; under each class: frames containing the class | class pixels.
Red columns = rare class (<= 3 videos or >= 50% of pixels in one video). Grey 0 = class absent from the video. (classes part 2/2, videos part 2/3)

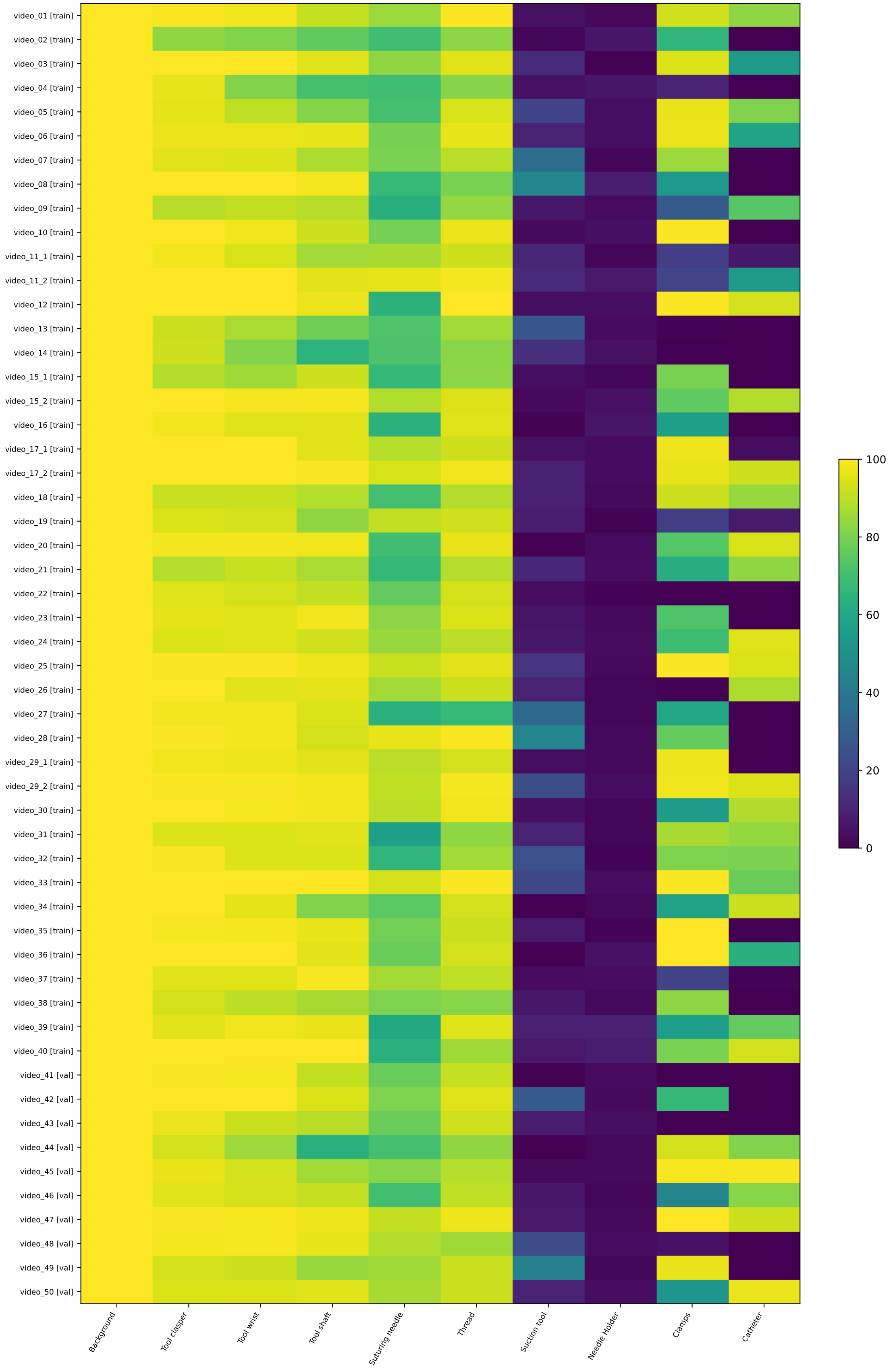
	Suction tool		Needle Holder		Clamps		Catheter	
Video [split]	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**
video_24 [train]	16	1.31M	7	673.2k	172	1.07M	238	2.02M
video_25 [train]	24	1.41M	4	580.2k	153	322.7k	146	1.66M
video_26 [train]	16	746.9k	2	327.0k	1	550	144	613.9k
video_27 [train]	64	5.10M	3	226.2k	113	31.9k	0	0
video_28 [train]	83	3.36M	4	668.5k	138	115.0k	0	0
video_29_1 [train]	10	2.32M	6	294.9k	251	369.7k	0	0
video_29_2 [train]	55	5.74M	8	934.2k	228	1.87M	221	1.89M
video_30 [train]	23	1.43M	9	1.38M	314	1.45M	505	5.14M
video_31 [train]	55	8.18M	8	791.6k	490	2.25M	475	14.24M
video_32 [train]	120	15.81M	4	376.5k	395	537.1k	394	4.95M
video_33 [train]	34	1.15M	5	184.9k	157	392.8k	122	331.9k
video_34 [train]	0	0	2	386.0k	58	98.9k	93	383.0k
video_35 [train]	36	2.10M	4	127.2k	504	2.18M	0	0
video_36 [train]	0	0	15	1.92M	339	5.78M	212	2.86M
video_37 [train]	11	498.4k	12	1.19M	74	100.9k	4	43.3k
video_38 [train]	20	399.5k	7	591.3k	254	527.6k	0	0
video_39 [train]	22	2.30M	22	1.77M	133	812.9k	182	2.30M
video_40 [train]	19	505.0k	21	3.46M	221	1.08M	257	425.4k
video_41 [val]	0	0	7	948.7k	0	0	0	0
video_42 [val]	103	7.11M	9	1.82M	238	1.26M	0	0
video_43 [val]	16	1.91M	7	1.89M	0	0	0	0
video_44 [val]	0	0	3	1.09M	133	229.5k	116	872.3k
video_45 [val]	3	209.0k	3	532.8k	116	633.4k	116	2.37M
video_46 [val]	42	6.52M	9	1.92M	287	557.9k	520	6.37M
video_47 [val]	23	1.08M	9	1.08M	334	3.05M	308	1.41M
video_48 [val]	71	8.91M	10	1.91M	15	68.1k	0	0

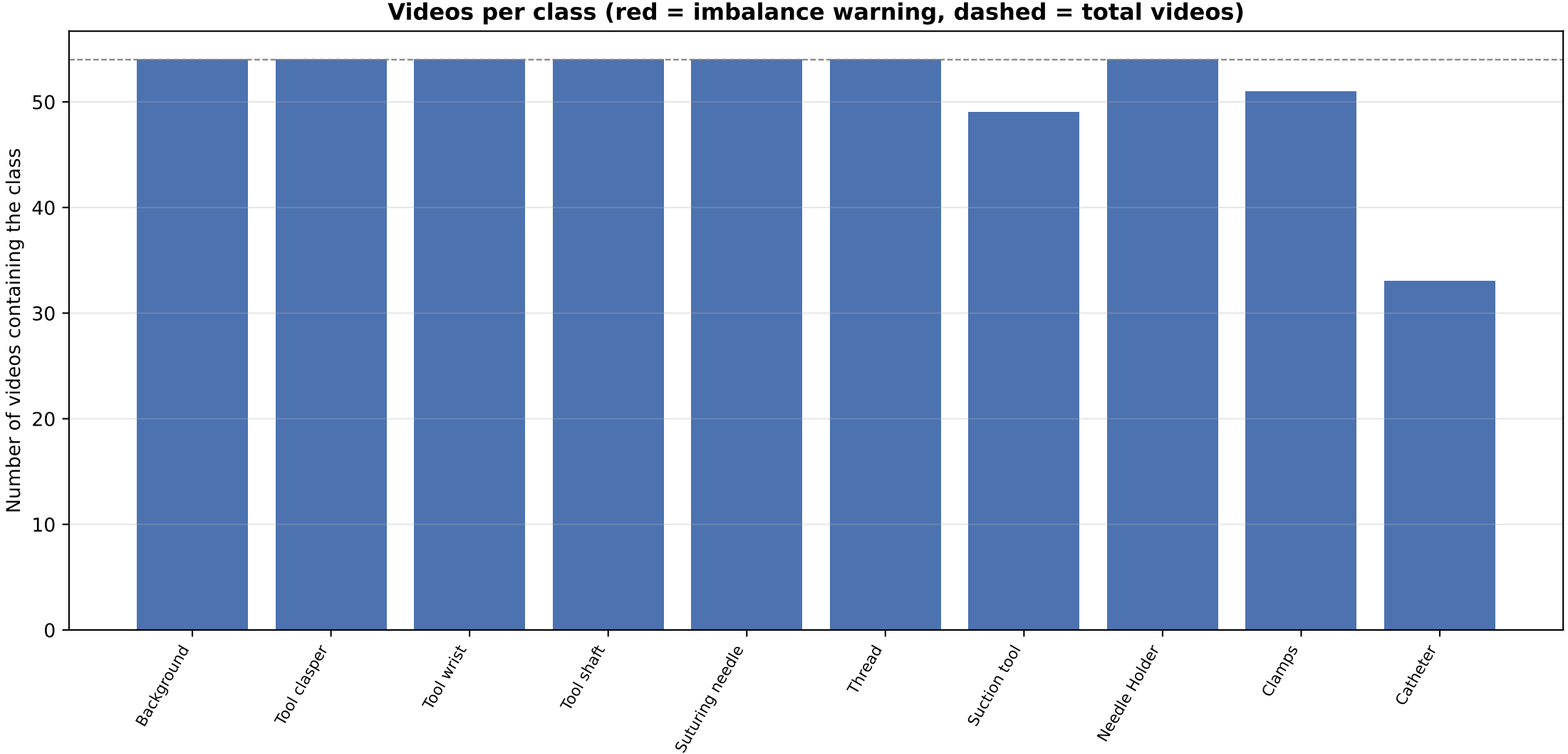
5. Video x Class table (frames and pixels)

Rows = videos, columns = classes; under each class: frames containing the class | class pixels.
Red columns = rare class (<= 3 videos or >= 50% of pixels in one video). Grey 0 = class absent from the video. (classes part 2/2, videos part 3/3)

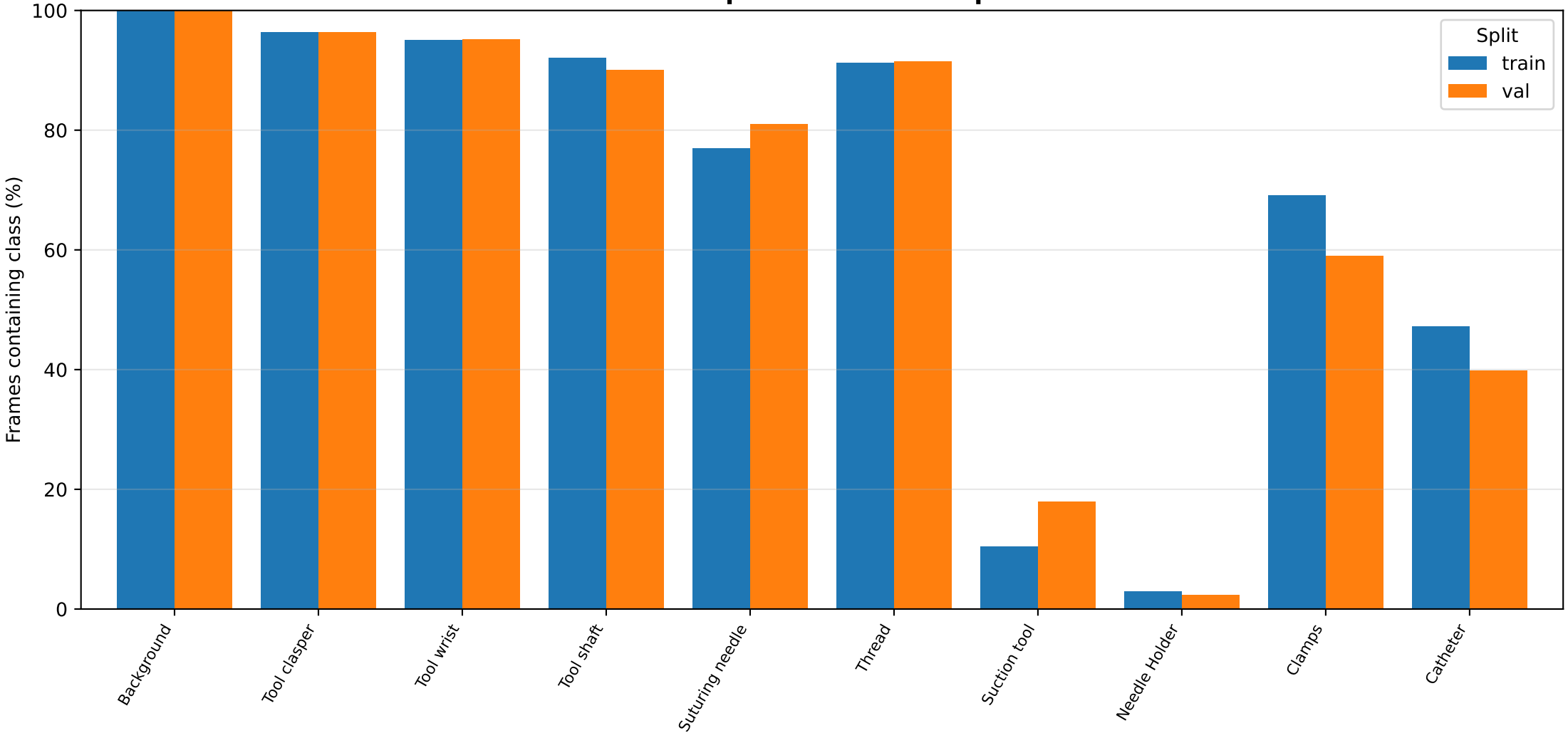
	Suction tool		Needle Holder		Clamps		Catheter	
Video [split]	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**	**frames**	**pixels**
video_49 [val]	301	47.70M	12	1.52M	665	5.21M	0	0
video_50 [val]	24	569.7k	8	579.5k	128	147.2k	236	4.08M
TOTAL	**1,942**	**212.59M**	**458**	**61.40M**	**10,927**	**59.83M**	**7,458**	**77.61M**

Class presence per video (% of the video's frames containing the class)

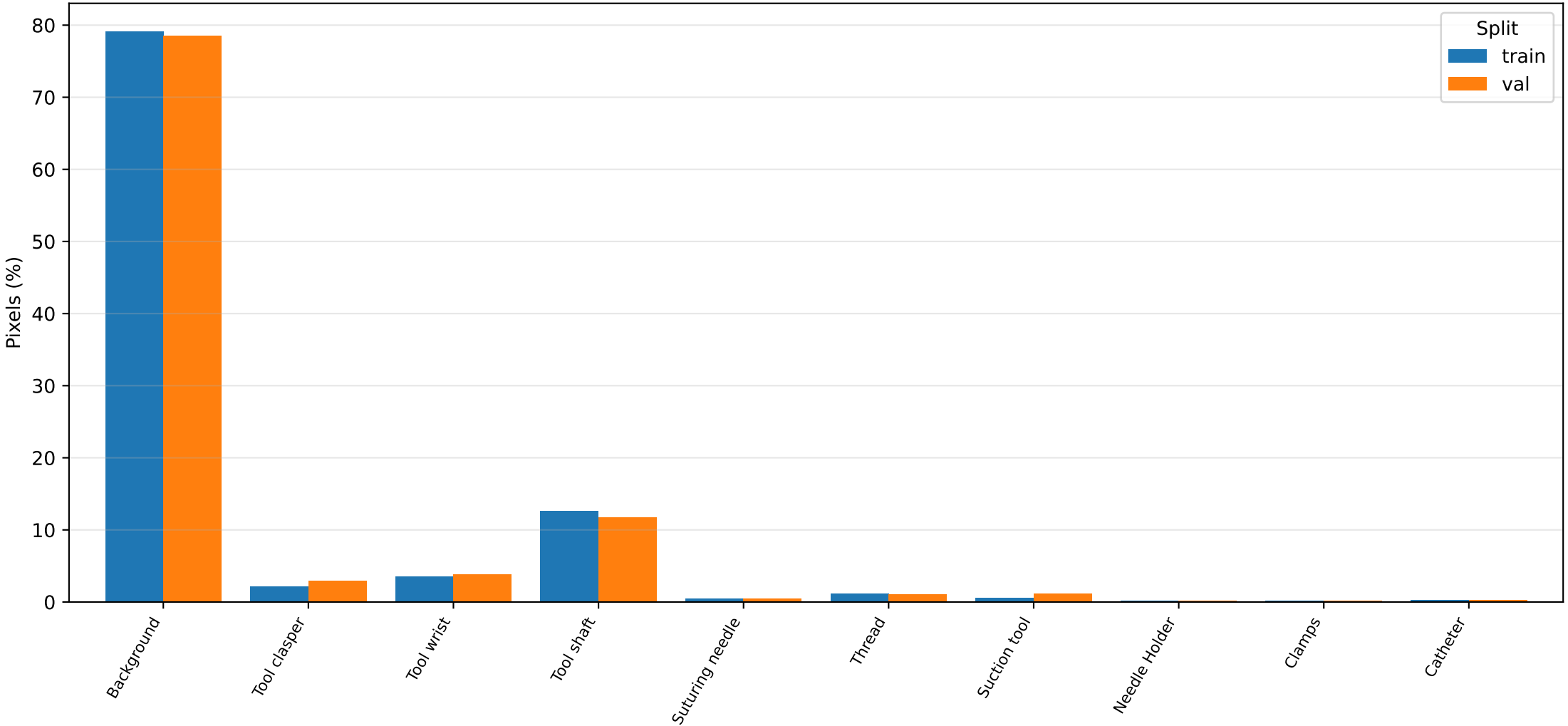



Videos per class (red = imbalance warning, dashed = total videos)
Number of videos containing the class
0
10
20
30
40
50
Background
Tool clasper
Tool wrist
Tool shaft
Suturing needle
Thread
Suction tool
Needle Holder
Clamps
Catheter

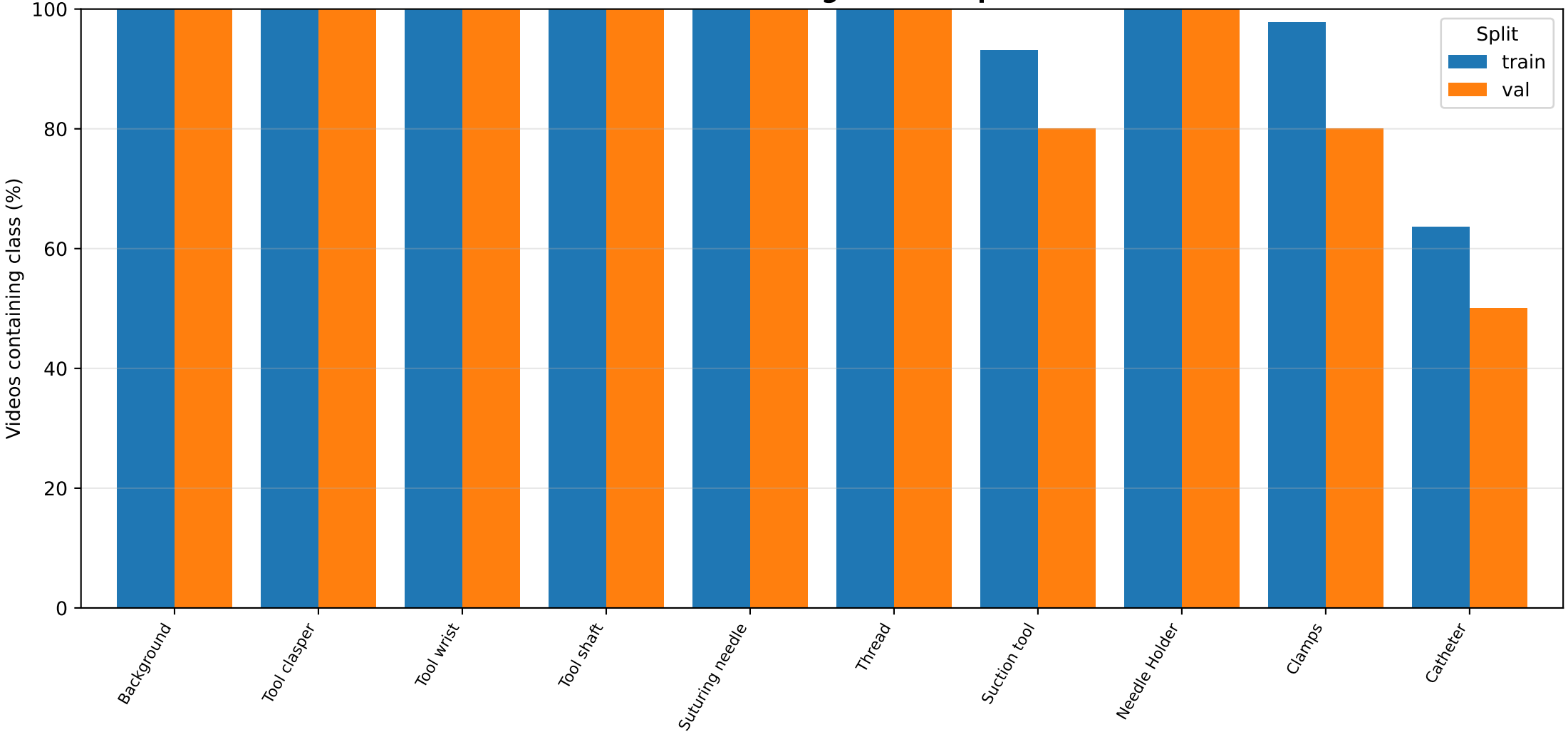
6. Class presence across splits



6. Pixel distribution across splits

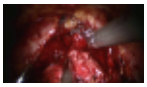


6. Video coverage across splits



7. Qualitative Analysis - train

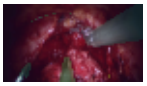
ge - #8535 (video_28)



Ground truth



Overlay



Classes: Background, Tool clasper, Tool wrist, Tool shaft, Suturing needle, Thread, Suction tool, Clamps

ge - #10091 (video_31)



Ground truth

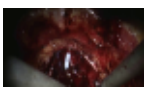


Overlay



Classes: Background, Tool clasper, Tool wrist, Tool shaft, Suturing needle, Thread, Clamps, Catheter

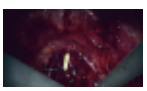
ge - #5647 (video_18)



Ground truth



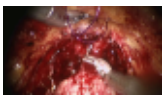
Overlay



Classes: Background, Tool clasper, Tool wrist, Tool shaft, Thread, Clamps, Catheter

7. Qualitative Analysis - train

ge - #11198 (video_35)



Ground truth

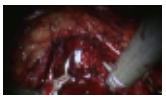


Overlay



Classes: Background, Tool clasper, Tool wrist, Tool shaft, Suturing needle, Thread, Clamps

age - #5723 (video_18)



Ground truth



Overlay



Classes: Background, Tool clasper, Tool wrist, Tool shaft, Suturing needle, Thread, Clamps, Catheter

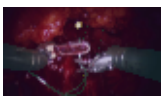
age - #1163 (video_03)



Ground truth



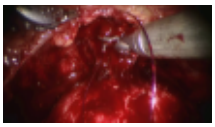
Overlay



Classes: Background, Tool clasper, Tool wrist, Tool shaft, Suturing needle, Thread, Clamps, Catheter

7. Qualitative Analysis - val

mage - #15169 (video_48)



Ground truth

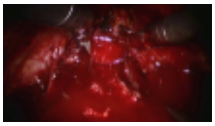


Overlay



Classes: Background, Tool clasper, Tool wrist, Tool shaft, Suturing needle, Thread

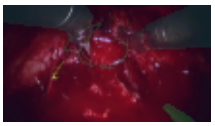
mage - #15556 (video_49)



Ground truth

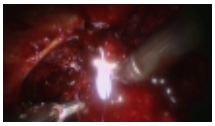


Overlay



Classes: Background, Tool clasper, Tool wrist, Tool shaft, Thread, Suction tool, Clamps

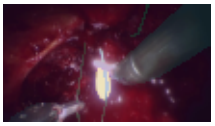
mage - #14450 (video_46)



Ground truth



Overlay



Classes: Background, Tool clasper, Tool wrist, Tool shaft, Suturing needle, Thread, Catheter

7. Qualitative Analysis - val

- #15835 (video_49)



Ground truth



Overlay



Classes: Background, Tool clasper, Tool wrist, Tool shaft, Suturing needle, Thread, Suction tool, Clamps

- #14469 (video_46)



Ground truth



Overlay



Classes: Background, Tool clasper, Tool wrist, Tool shaft, Suturing needle, Thread, Suction tool, Clamps, Catheter

- #13332 (video_42)



Ground truth



Overlay



Classes: Background, Tool clasper, Tool wrist, Tool shaft, Suturing needle, Thread, Clamps

8. Conclusions

Dataset: sarrarp50

Frames, pixels and videos measure three different things:
- Frames: how often the class occurs across images.
- Pixels: how much image area the class occupies.
- Videos: how many independent patients/procedures show the class.

A class present in many frames but in only a few videos is a weak basis for generalisation: frames of the same video are highly correlated, so the effective sample size is closer to the number of videos than to the number of frames.

Splits are made at video level, so a rare class can end up entirely in train or entirely in validation/test. Check 'Classes missing from a split' in the executive summary before interpreting per-class metrics.

Classes flagged in red (few videos, or concentrated in one video) deserve special care: per-class metrics on them are high-variance, and class weighting or oversampling should be evaluated.